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A Critical Value

What Is The Real Meaning Of The EMA Alpha?

In an exponential moving average, the smoothing factor, also known as a weighted multiplier or alpha, is the parameter that determines how values in the lookback period are weighted. Weights decline exponentially as the data points get older. How is alpha found, and why is this significant? Here are some analytical and practical insights.

by John F. Ehlers

An exponential moving average (EMA) is a first-order smoothing filter. That is, it only uses one previous result in computing the current value. It is expressed by the equation:

$$Output = \alpha * Input + (1 - \alpha) * Output[1]$$

where output[1] is the EMA computed value one bar ago.

The equation is dominated by the value of alpha (α), so alpha is important. But what does it mean when applied to trading data?

In his May 1984 article "Moving Averages Versus Exponential Moving Averages" in this magazine, Jack Hutson related alpha to the length of a simple moving average (SMA) on the basis of equivalent lag. He showed the relationship as:

$$\alpha = 2 / (N + 1)$$

where N is the length of the equivalent SMA.

The EMA is a smoothing filter, and therefore is a lowpass filter. That is, it passes the low-frequency (long wavelength) components in the data spectrum and attenuates the high-frequency components. The distinction between the low-frequency components and high-frequency components is a bit fuzzy, but the critical period is generally defined as the *period*, where the amplitude of the filter output has an amplitude of –3 dB relative to the filter input. So the real meaning of alpha lies in its relationship to the critical period.

The derivation of alpha in terms of the critical period of the filter gets a bit mathy because the use of complex variables is required. So, if you are not familiar with Z transforms and complex variables, just skip the following explanation.

We can rearrange the original EMA equation to be:

$$\text{Output} - (1 - \alpha)*\text{Output}[1] = \alpha *\text{Input}$$

The transfer response of a filter is defined as the ratio of its input to its output. So, the transfer response of an EMA in terms of Z transforms becomes:

$$\text{Transfer} = \alpha / (1 - (1 - \alpha)*Z^{-1})$$

To find the frequency response of the filter, we can substitute $e^{-j(wt)}$ for Z^{-1}. And:

$$e^{-j(\omega t)} = \text{Cos}(\theta) - j\text{Sin}(\theta)$$

where $\theta = \omega t$

Making these substitutions, the EMA transfer response is:

$$\text{Transfer} = \alpha / (1 - (1 - \alpha)*\text{Cos}(\theta) - j(1 - \alpha)*\text{Sin}(\theta))$$

We multiply the numerator and denominator by the complex conjugate of the denominator to place the real and imaginary components solely in the numerator. The −3 dB point of the transfer response occurs when the real and imaginary components are equal, giving a 45-degree phase shift. Equating the real and imaginary components, and ignoring the constants, we get:

$$1 - (1 - \alpha)*\text{Cos}(\theta) = (1 - \alpha)*\text{Sin}(\theta)$$

The final result is obtained by rearranging the equation to isolate alpha, to get:

$$\alpha = (\text{Cos}(\theta) + \text{Sin}(\theta) - 1) / (\text{Cos}(\theta) + \text{Sin}(\theta))$$

where $\theta = 360 / \text{Period}$ in degrees or
$\theta = 2*\pi / \text{Period}$ in radians.

The relationship between the critical period of the filter and the EMA alpha is reasonably complex. An approximation that's easier to remember is:

$$\alpha = 4*\pi / (2*\pi + \text{Period})$$

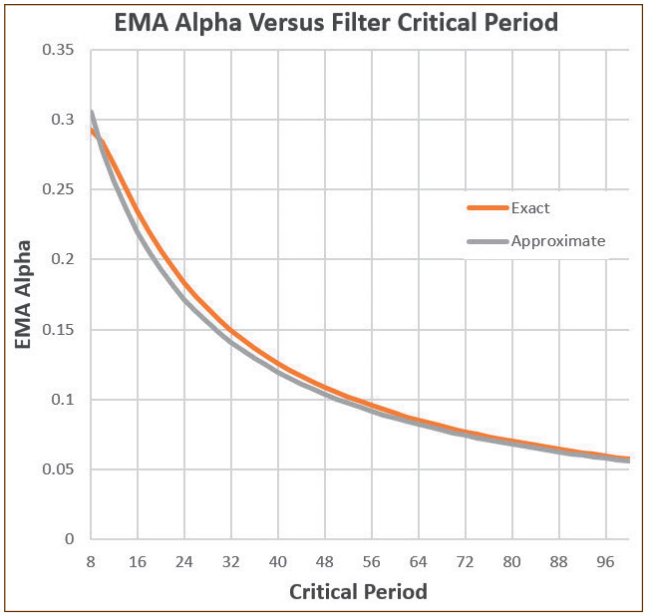

FIGURE 1: EMA ALPHA AND FILTER CRITICAL PERIOD. The relationship between EMA alpha and filter critical period is nonlinear.

The exact relation between alpha and period is shown in orange and the approximation is shown in blue in Figure 1.

Note that the critical period stops at 8 in Figure 1. An eight-bar period is about the shortest period you want to use in technical analysis because of aliasing. The data has a statistically pink spectrum, where the cycle component doubles for each doubling of the cycle period. Since the sampling rate is once per bar, the shortest possible period to use is the Nyquist rate of a two-bar cycle. Shorter cycle components are not observable, but are folded back into the observable spectrum.

The pink spectrum shape holds for cycle periods shorter than Nyquist. Therefore, the data for a one-bar cycle component folds back with an amplitude of 0.5, a half-bar cycle component folds back with an amplitude of 0.25, etc. The generalized expression for the alias amplitude as a function of the observable cycle period is 1 / (2*Period). Therefore, the aliasing error does not become sufficiently small enough to be ignored until the observed cycle period is 8 bars or longer.

Alpha is important. But what does it mean when applied to trading data?

Conclusion

The alpha of an EMA can be computed from its critical period as $\alpha = 4*\pi / (2*\pi + \text{Period})$.

However, the EMA is only a first-order IIR filter. Better filtering results can be obtained by using second-order filters such as the SuperSmoother or UltimateSmoother (see the "Further reading" section for my April 2024 article on the ultimate smoother).

It is good practice to use cycle periods longer than an eight-bar period in technical analysis due to aliasing.

John Ehlers is a retired electrical engineer and a retired technical analyst, specializing in the application of DSP (digital signal processing) to trading. For more information, see www.mesasoftware.com.

Further reading

Jack K. Hutson [1984]. "Moving Averages Versus Exponential Moving Averages," *Technical Analysis of* STOCKS & COMMODITIES, Volume 2: May.

Ehlers, John F. [2024]. "The Ultimate Smoother," *Technical Analysis of* STOCKS & COMMODITIES, Volume 42: April.


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